



Published in final edited form as:

Environ Int. 2026 May ; 211: 110260. doi:10.1016/j.envint.2026.110260.

Associations between self-reported personal care products use and menstrual cycle length and regularity in a US digital cohort

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Disclosure statement

K.A. and S.H.S. own Apple Inc. stock and are employed by Apple Inc. Other co-authors have no conflict of interest. There were no financial relationships with any organizations that might have an interest in the submitted work in the previous 3 years and no other relationships or activities that could appear to have influenced the submitted work.

Trial registration

This study is not a clinical trial. However, the parent cohort, Apple Women's Health Study, is registered on [Clinicaltrials.gov](https://clinicaltrials.gov/ct2/show/NCT04196595) (NCT04196595): <https://classic.clinicaltrials.gov/ct2/show/NCT04196595>.

Abstract

Background: Personal care products (PCPs) may contain endocrine-disrupting chemicals (EDCs) that can impact menstrual health. Despite widespread usage, little is known about the associations of PCP usage, EDC avoidance, and menstrual cycle characteristics.

Methods: This analysis included female participants from a US-based digital cohort who enrolled between 11/2019–04/2025, provided consent, aged < 50 years, completed PCP questions (past year usage frequency, age at initiation, and ingredient avoidance when purchasing PCPs), and either reported their usual menstrual cycle length (<21, 21–39, ≥40 days/too irregular) (n = 4,155) or shared logged cycle data (n = 4,482 with 61,079 cycles). We used confounder-adjusted multinomial logistic regression or linear mixed effect models to examine associations between each exposure and cycle length or variability.

Results: PCP usage frequency varied considerably by product (e.g., 54%, 8%, 2% using body fragrance, dry shampoo, and self-tanner ≥ 3 times/week; 23% and 2% using permanent and semi-permanent hair dye ≥ 4 times/year and ≥ monthly). We found consistent associations between both frequent current usage and childhood initiation of the aforementioned PCPs with very long, very short, or irregular cycles. Participants avoiding harmful ingredients (petroleum propellants, formaldehyde, lead, mercury, or talc) were more likely to report regular cycles (lower cycle variability). Effect modifications were found, and individuals who self-reported fibroids or endometriosis diagnoses or of younger age (18–29 years) were more susceptible to some of these associations.

Discussion: These findings suggest that EDC-associated PCPs may impact reproductive health, thus suggesting PCP use as an add to modifiable environmental factors affecting menstrual health worthy of future research.

Keywords

Endocrine disrupting chemicals; Personal care products; Metals; Menstrual cycle; Irregular menses; Menstrual health

1. Introduction

Approximately 1.8 billion individuals experience monthly menstruation globally (Ramaiyer et al., 2023). Menstrual cycle characteristics (MCCs), such as cycle length and regularity, serve as vital signs from menarche through menopause (Rosen Vollmar et al., 2025). The International Federation of Gynecology and Obstetrics (FIGO) indicates that menstrual cycles occurring every 24–38 days are typically associated with ovulation, whereas menstrual bleeding associated with ovulatory disorders is typically irregular in timing (Munro et al., 2018). Studies found that individuals with menstrual cycle lengths (MCL) of < 21 days (very short) or ≥ 35/≥40 days had a higher risk of cardiometabolic conditions (Huang et al., 2023; Solomon et al., 2001). Abnormal or irregular cycles during reproductive years are also associated with other adverse health outcomes, such as infertility, cancers, pregnancy complications (e.g., gestational diabetes, preeclampsia, preterm birth), and menopausal symptoms (Abetew et al., 2011; Bonnesen et al., 2016; Cao et al., 2024; Cirillo et al., 2016; Mínguez-Alarcón et al., 2022; Schmeler et al., 2005; Soria-Contreras

et al., 2022; Wang et al., 2024). Globally, menstrual irregularity impacts around 10–30% of menstruators with variations by geographical locations and demographics (Harlow and Campbell, 2004; Nobles et al., 2022; Whitaker and Critchley, 2016), contributing to substantial healthcare burdens. Understanding factors that affect MCCs, such as modifiable environmental exposures, is therefore critical for health promotion over the life course.

Emerging evidence suggests that exposure to endocrine disrupting chemicals (EDCs) and heavy metals may impact cycle length or regularity, by altering the balance of hormones in the hypothalamic-pituitary-ovarian (HPO) axis that regulate menstrual cycles (Buck Louis et al., 2011; Ding et al., 2022; Hammer et al., 2020; Jackson et al., 2011; Jukic et al., 2016; Li et al., 2024b; Lum et al., 2017; Lyngsø et al., 2014; Nishihama et al., 2016; Pollack et al., 2018; Zhu et al., 2019). These studies have found high exposure levels of phenols, parabens, phthalates, triclosan, per- and polyfluoroalkyl substances (PFAS), polychlorinated biphenyls (PCBs), or mercury to be associated with either shorter, longer, or more irregular cycles. However, findings were not always consistent across studies, and were often limited by relatively small to moderate sample sizes and reliant on one-time exposure biomarker assessment that may be subject to potential reverse causation, since menstrual blood is an excretion route for some of these chemicals (Silva et al., 2022; Upson et al., 2022; Wu et al., 2015). Additionally, some studies excluded participants with known fertility or gynecological conditions, and these findings may not fully represent individuals with the most severe menstrual disorders (e.g., very irregular or very short/long cycles) often indicating underlying reproductive disorders.

Personal care products (PCPs) are common sources of exposure to these endocrine disruptors including phthalates, parabens, PFAS, triclosan, chemical fragrances, lead, or mercury (Bloom et al., 2024; Braun et al., 2014; Dodson et al., 2012; Johnson et al., 2022; Whitehead et al., 2021). These products can lead to exposure through dermal absorption and inhalation. Despite widespread use of PCPs, limited research has directly examined associations between PCP use and menstrual health among reproductive-aged females. Understanding the direct associations between PCP usage patterns, EDC awareness, and MCCs can provide novel insights into modifiable factors for menstrual health. Additionally, examining current and childhood/adolescence PCP usage patterns provides further insights into critical exposure windows (Sommer et al., 2022) to inform public health recommendations.

In this study, we evaluated associations of self-reported PCP usage (early-life and current) and avoidance of PCP ingredients with cycle length and cycle variability. We analyzed data from both participant self-reported MCL and cycle logging smartphone application data within a United States (US) digital cohort.

2. Methods

2.1. Study design and Population

The Apple Women's Health Study (AWHS) is a prospective, digital cohort study. Eligible participants—those who have menstruated at least once, live in the United States, are at least 18 years old (19 in Alabama and Nebraska, 21 in Puerto Rico), are able to communicate in

English, are the sole users of their iCloud account, and provide written informed consent at enrollment—enrolled using the Apple Research app on their iPhone. Recruitment began on 11/14/2019 and is ongoing. Further details on the AWHs study design, eligibility, and data were described previously (Mahalingaiah et al., 2022). The Institutional Review Board at Advarra approved this study (CIRB #PRO00037562).

For this analysis, we included female participants who enrolled between November 14, 2019 and April 1, 2025, responded to survey questions on self-reported PCP usage (details described in Table S1) and reproductive, sociodemographic, medical, and lifestyle characteristics (surveys distributed at enrollment and updated every 12 or 18 months). We used first responses to each recurrent survey for this analysis. We excluded participants who reported current hormone use; were pregnant, breastfeeding, or menopausal at baseline; or who reported a history of hysterectomy. Participants aged ≥ 50 years were also excluded as previous studies in this cohort indicated highly irregular cycles for this age group (Li et al., 2024a; Li et al., 2023), likely influenced by perimenopause (Appiah et al., 2021).

We used two analytical datasets for our analyses: (1) **Dataset #1 (n = 4,155 participants)** for PCP usage, EDC awareness, and *survey-based* MCL outcomes and (2) **Dataset #2 (n = 4,482 participants with a total of 61,079 eligible logged menstrual cycles)** for PCP usage, EDC awareness, and *app data-based* MCL outcomes. The different methods for reporting MCL outcomes are detailed below. Further details of participants included in these two analytical datasets can be found in Figs. S1. Fig. S2 shows the number of overlapped participants in both datasets (n = 2,240).

2.2. PCP usage

Participants were asked to respond to a series of questions related to their usage of a variety of PCPs, including make up (blush/bronzer, eye makeup products including eyeliner, eyeshadow, eyeshadow primer, eyebrow products, or mascara, lip products, foundation/concealer), body fragrance, nail products (nail polish and nail polish remover), skincare products (cleansers, eye treatments, facial treatments, moisturizer, petroleum jelly, skin lightener, anti-aging products, deodorants, sun care products, and self-tanner), and hair care products (rinse out conditioners, shampoo, dry shampoo, permanent dye/coloring, semi-permanent dye/coloring, touch up hair coloring products, hair growth products, leave in conditioner, perms/relaxers, oils, gels, and smoothing/keratin products). Participants were asked to select if they have ever used each of these PCPs. For each PCP selected, participants were further asked their usage frequency in the past year (options for most PCPs include: *more than once a day, 7 days/week, 3–6 days/week, 1–2 days/week, a few times a month, a few times a year or less, never (in the past year)*); alternatively, specific options were given for certain permanent or semi-permanent hair products, including permanent dye/coloring or perms/relaxers [*more than once a month, about once a month, about every 2 months, 4–5 times/year, 3 times/year or less, never (in the past year)*], and semi-permanent dye/coloring [*every two weeks, a few times a month, a few times a year or less, never (in the past year)*] and at what age they began using the product (≤ 9 , 10–12, 13–18, 19–25, 26–30, 31–38, 39–45, 46–51, ≥ 52 years). For each hair product, participants were further asked if the product typically has a fragrance. Relevant survey questions are available in

Supplemental Table S1. Those who indicated “I don’t know” or “I prefer not to answer” for each PCP variable were excluded from the relevant analyses.

In our analyses, we grouped **usage frequency in the past year** into three-level categorical variables: (1) for most PCPs: *never in the past year to a few/year* (referent), *a few/month to 1–2 days/week*, and ≥ 3 days/week; (2) for permanent dye/coloring or perms/relaxers: *never in the past year* (referent), *1–3 times/year*, and ≥ 4 times/year; (3) for semi-permanent dye/coloring: *never in the past year* (referent), *a few times/year*, and *a few times/month or more*. **Age at initiation** was also grouped into a three-level categorical variable: *age ≥ 19 or never used* (referent), *age 13–18*, and *age ≤ 12 years*. We recognize that for some PCPs with patterns of uptake later in adulthood (e.g., anti-aging, hair growth, or eye treatment products), some participants who report never having used a product at the time of survey response (shown in Fig. 1) may still initiate later in life. As all participants in this analytical dataset were at least 18 years old (19 in Alabama and Nebraska, 21 in Puerto Rico), by aggregating never users with those who initiated use at age ≥ 19 , we create a referent category of individuals who have avoided exposure during key early developmental periods like childhood/adolescence (periods that may be associated with higher vulnerability to endocrine disruption). This approach acknowledges the uncertainty of “never” use among younger participants, and aligns with our aim to estimate risks attributable to early (childhood/adolescence) initiation.

Among hair products users, we evaluated whether the hair products that they used “have a fragrance” or not as a secondary exposure variable. This question (related to whether products contained fragrance) was asked only for hair products.

2.3. EDC awareness

Participants were asked to select all that apply from a list of product ingredients that they try to avoid, including aerosol sprays with petroleum propellants, butylated hydroxyanisole (BHA), coal tar dyes, diethanolamine (DEA), formaldehyde, fragrance, isobutane, propane and other propellants, lead, mercury, parabens, phthalates, polyethylene glycols (PEG), per- and polyfluoroalkyl substances (PFAS), talc, toluene, products with a long list of chemicals, or the exclusive choices of “none of the above” or “I don’t know”. These responses are best interpreted as indicators of awareness and intention to avoid chemicals of concern, rather than precise measures of successful avoidance or direct assessments of chemical concentrations in the body. Those who indicated “I prefer not to answer” were excluded from the analysis. We evaluated *avoiding* (vs. *not avoiding*) each ingredient individually as well as grouping into a categorical variable of total number of ingredients avoided: 0 (referent), 1 to 5, 6 to 10, and ≥ 11 ingredients. For EDC avoidance, we also evaluated whether those who reported avoiding ingredients were different from the others.

2.4. Menstrual cycle characteristics (MCCs)

For Dataset #1, eligible participants responded to the question, “In the last year, how many days did your typical menstrual cycle last?” with answer options of *< 21 days*, *21–25 days*, *26–31 days*, *32–39 days*, *40–50 days*, *51 + days*, or *too irregular to estimate*. We further grouped self-reported MCL as: 1) very short: *<21 days* (Bull et al., 2019; Whooten et al.,

2024); 2) 21–39 days (as referent level for multinomial logistic regression); and 3) very long (≥ 40 days) or too irregular to estimate. While the typical cutoff value for long MCL is 35 days (Bull et al., 2019; Soria-Contreras et al., 2022), due to the categorical options available to participants in this cohort, we used the cutoff of ≥ 40 days as very long cycles (considered at the tail of common MCL distributions).

For Dataset #2, cycle length and variability were derived from participant logged cycles. Similar to previous work, length (in days) for each cycle was calculated by subtracting the reported date of first bleeding from the subsequent reported first bleed date, and ongoing cycles were excluded from our analyses. As shown in Fig. S1, we restricted to participants who logged ≥ 3 eligible menstrual cycles using the Cycle Tracking feature on the Apple Health app or any third-party application that the participant allows to write to the Health app. An eligible cycle for this analysis was defined as a cycle between 10–90 days long for which the participant confirmed no hormonal use, pregnancy, or breastfeeding. Atypically long cycles based on individual thresholds (3% of all logged cycles) were also excluded using methods described previously (in short, we used an individual-specific threshold, factoring in both typical MCL and cycle variability of each individual) (Li et al., 2024a; Li et al., 2023). Eligible cycles within each participant were used as a longitudinal outcome in the linear mixed effects models described below.

2.5. Covariates

The following self-reported sociodemographic, reproductive, medical, and lifestyle characteristics were included as covariates that were adjusted for in our regression models, based on *a priori* knowledge on factors relevant to the exposure and outcome: (1) age in years (calculated from birth year and enrollment year); (2) race/ethnicity (Asian, Hispanic, non-Hispanic Black, non-Hispanic White, or multiple races [self-identified > 1 option]/other races [American Indian or Alaska Native, Middle Eastern or North African, Native Hawaiian or Pacific Islander, or none of these categories can fully describe the participant]); (3) subjective socioeconomic status (SES), assessed using the MacArthur Scale of Subjective Social Status where participants self-rank themselves on a social ladder relative to others (grouped as 0–3 [low], 4–5 [medium], 6–9 [high]) (Moss et al., 2023); (4) education ($\leq$ high school, some college or technical school, college, or graduate degree); (5) body mass index (BMI), calculated from self-reported weight and height at enrollment (categorized as < 18.5, 18.5–24.9, 25.0–29.9, 30.0–34.9, 35.0–39.9, or ≥ 40 kg/m²) (CDC, 2024); (6) number of reported pregnancies that were ≥ 24 weeks (grouped as 0, 1, 2+); (7) typical exercise minutes per week, defined as any moderate to vigorous leisure time activity (grouped as none, 1–75, 76–150, 151–300, or >300 min/week); (8) dietary patterns including low calorie/carb/fat (yes/no), high fat (yes/no), high protein (yes/no), or vegetarian/vegan (yes/no); (9) the Perceived Stress Scale 4 (PSS-4) derived from participant's response to the four relevant questions (Cohen et al., 1983), with a total PSS-4 score (range 0–16); (10) typical weekday and weekend sleep hours per night, derived from reported time falling asleep and waking up; (11) alcohol consumption (grouped as no/light alcohol consumption, light consistent/rare binge drinker, or moderate-to-heavy consistent drinker/frequent binge drinker based on The National Institute on Alcohol Abuse and Alcoholism's [NIAAA] definition of heavy drinking (Creazzo, 2022)); (12) tobacco smoke (grouped as never, past, or current);

(13) current e-cigarette use (none, some days, everyday); and (13) current marijuana use (none, some days, everyday). The following reproductive conditions were evaluated as potential effect modifiers: self-reported physician diagnosis of polycystic ovary syndrome (PCOS), uterine fibroids, or endometriosis (yes/no).

2.6. Statistical analysis

In Dataset #1, for the associations between each PCP variable and self-reported MCL from survey response, we used multinomial logistic regression to estimate the odds ratios (ORs) and 95% confidence intervals (CIs) of very short cycles (<21 vs. 21–39 days) and very long cycles (≥ 40 days or too irregular to estimate vs. 21–39 days). In Dataset #2, for the associations between each PCP variable and mean MCL from logged cycles, we used linear mixed effect (LME) models with random participant-specific intercepts to estimate the differences and 95% CIs in MCL by changes in PCP frequency, age at initiation, or EDC awareness categories. For each association, we utilized (1) a Model 1 that only adjusted for age, race/ethnicity, and SES, and (2) a fully adjusted Model 2 that adjusted for all aforementioned covariates. To quantify the associations between each PCP variable and within-individual cycle variability (a measure of cycle irregularity), we constructed log-linear models for residual variance in the adjusted LME models to estimate the mean within-individual standard deviations (SDs) of cycles (in days) and 95% CIs in each PCP frequency, age at initiation, or EDC awareness category, similar to previous work within this cohort (Asokan et al., 2024; Li et al., 2023).

As secondary analyses, we evaluated potential effect modifications for the associations of current PCP usage frequency or EDC avoidance with MCCs by the following factors: 1) self-reported PCOS, uterine fibroids, endometriosis diagnosis (yes vs. no), as these conditions may have altered physiology (e.g., hormone disruption or inflammatory responses) (Bariani et al., 2020; Stephens et al., 2022; Velez et al., 2021) that could potentially modify the impact of PCPs/EDCs on the risk and severity of cycle irregularity; 2) age categories (18–29 or 40–49 vs. 30–39), to test if any age-specific biological responses to EDCs are present for MCCs (Pollack et al., 2022); and 3) race and ethnicity (non-Hispanic Black, Hispanic, Asian, or multiple/other races vs. non-Hispanic White), to explore whether any systemic factors may impact both exposure levels and differential susceptibility (James-Todd et al., 2016). Potential effect modification was estimated by including an interaction term between each exposure and modifier and statistical significance estimated with Wald tests. Significant interactions were further evaluated via evaluating the strengths of exposure-outcome associations in subgroups stratified by the significant effect modifiers. Additionally, among hair products users, we further evaluated the associations of whether the hair products that the participants typically use contained fragrance or not (yes vs. no) and MCCs. Furthermore, to characterize potential co-exposure patterns, we first constructed heatmaps of pairwise Cramer's V for past year frequency of all PCPs and age at initiation of all PCPs, respectively. We then applied: 1) a model-based clustering algorithm with feature selection (Bellavia et al., 2025; Marbac et al., 2020; Marbac and Sedki, 2019) to derive class memberships of participants reflecting overall PCP use patterns, and examined the associations between these dimension-reduced class memberships and MCCs; and 2)

penalized regression models using Elastic Net to evaluate products retained in the models and explore product-specific associations while accounting for co-exposures.

As sensitivity analyses to account for varying state regulations regarding EDCs in PCPs, we categorized participants' current states of residence into regions based on historical and current state-level legislations as detailed in Supplemental Table S2 (U.S. GAO Office, 2023), and we evaluated associations between current PCP use frequency and MCCs stratified by these regional categories. We also evaluated the prevalence of actively attempting to conceive based on participant's self-report for their monthly logged cycles.

Analyses were conducted in Python (Python Software Foundation, version 3.6) and R (R Project for Statistical Computing, version 4.1.2). All statistical tests were two-sided with 95% CIs. In the main analyses, p-value cutoffs of 0.05 were used to determine statistical significance. To evaluate the robustness of findings while accounting for multiple comparisons, we conducted a sensitivity analysis using the Benjamini-Hochberg approach to control for false discovery rate (FDR) for all main exposure-outcome associations. As further interpretations in this sensitivity analyses, FDR-adjusted p-values ≤ 0.05 were considered robust, and FDR-adjusted p-values ≤ 0.10 were considered exploratory/suggestive for future research.

3. Results

3.1. Baseline characteristics and exposure distributions

Baseline characteristics of the 4,155 AWHs participants with relevant survey-based MCL data and 4,482 participants with relevant logged cycle data are shown in Table 1 and Table S2. Among the 4,155 participants with self-reported MCL information from survey response (Dataset #1), median (IQR) age at enrollment was 34 (27–40) years. The majority self-evaluated as in the medium–high SES categories (71%). Around 33% had a BMI of 18.5–24.9 kg/m². About 29% self-reported exercising > 150 min/week (Table S2). Compared to Dataset #1, the 4,482 participants with 61,079 logged cycles (per person range: 3–83; per person median [IQR]: 9 [5 – 18] cycles) from mobile applications (Dataset #2) were older by approximately 1 year (median [IQR]: 35 [28–41]) and reported a higher percentage of medium–high SES (76%). Notably, 12–15%, 6%, and 5–6% of participants in either dataset reported a PCOS, fibroids, or endometriosis diagnosis, respectively (Table S2); when stratified by self-reported MCL, compared to those with MCL of 21–39 days, those with very long or irregular MCL had higher prevalence of PCOS (31% vs. 12%), while those with very short MCL had higher prevalence of fibroids (8% vs. 6%).

Distributions of self-reported ever usage of each PCP are shown in Fig. S3. In summary, some products like shampoo (99%), deodorant (96%), and moisturizer (94%) have been used by most participants at least once in their life, while some products like petroleum jelly (45%), hair perms/relaxers (19%), self-tanner (33%), skin lightener (5%) have been used by fewer participants. Distributions of usage frequency (Fig. 1A) and age at initiation (Fig. 1B) for selected PCPs in Dataset #1 are shown in Fig. 1 (full distributions of all PCPs in Datasets #1 and #2 are available in Supplemental Tables S3 and S4). Percentages (95% CIs) of product ingredient avoidance in Datasets #1 and #2 are summarized in Fig. 2. In Dataset

#1, among participants who provided a response other than “I don’t know”, around 19%, 41%, 23%, and 17% reported avoiding 0, 1 to 5, 6 to 10, and ≥ 11 ingredients, respectively (similarly in Dataset #2, these percentages were 19%, 39%, 25%, and 17%).

3.2. Current PCP usage frequency and MCCs

Fully adjusted estimates (Model #2) of current usage frequency of select PCPs with MCL categories in Dataset #1 and mean MCL and within-individual cycle variability in Dataset #2 are summarized in Fig. 3 (effect estimates for all PCPs in both datasets and both Models #1/#2 are available in Supplemental Table S5). Using perms/relaxers ≥ 4 times/year (vs. never in the past year) was associated with 2.63 (95% CI: 0.75, 9.26) times higher odds of very short cycles; using hair gel ≥ 3 days/week (vs. never in the past year to a few/year) or permanent dye/coloring products ≥ 4 times/year (vs. never in the past year) was associated with 1.63 (95% CI: 1.05, 2.54) times and 1.38 (95% CI: 1.04, 1.84) times higher odds of very long/irregular cycles; using body fragrance ≥ 3 days/week yielded a 6% higher cycle variability compared to never in the past year to a few/year [estimated within-individual SD: 4.84 (95% CI: 4.75, 4.93) days vs. 4.58 (95% CI: 4.51, 4.64) days]; using dry shampoo ≥ 3 days/week yielded a 12% higher cycle variability compared to never in the past year to a few/year [estimated within-individual SD: 5.39 (95% CI: 5.18, 5.61) days vs. 4.83 (95% CI: 4.79, 4.88) days]. Of note, using nail polish ≥ 3 days/week (which can include re-application of top coats) vs. never in the past year to a few/year was associated with both 1.76 (95% CI: 1.04, 2.98) times higher odds of very short cycles and 1.59 (95% CI: 1.05, 2.42) times higher odds of very long/irregular cycles, as well as yielding a 19% higher cycle variability [estimated within-individual SD: 5.51 (95% CI: 5.18, 5.83) days vs. 4.63 (95% CI: 4.59, 4.68) days].

3.3. PCP age at initiation and MCCs

Fully adjusted estimates (Model #2) of age at initiation of select PCPs with MCL categories in Dataset #1 and mean MCL and within-individual cycle variability in Dataset #2 are summarized in Fig. 4 (effect estimates for all PCPs in both datasets and both Models #1/#2 are available in Supplemental Table S6). In Dataset #1, the strongest association was observed with initiating semi-permanent dye/coloring at salons ≤ 12 years old (vs. ≥ 19 or never), resulting in a 2.49 (95% CI: 1.23, 5.02) times higher odds of very long/irregular cycles. In Dataset #2, those who initiated using dry shampoo during ≤ 12 or 13–18 years old had a 30% or 17% higher cycle variability compared to those who initiated ≥ 19 years old or never used dry shampoo [estimated within-individual SD: 6.16 (95% CI: 5.09, 7.23) days, 5.55 (95% CI: 5.34, 5.76) days, vs. 4.73 (95% CI: 4.69, 4.76) days]. Those who initiated using permanent dye/coloring at home ≤ 12 years old also yielded a 24% higher cycle variability than those who initiated ≥ 19 years old or never used this product [estimated within-individual SD: 5.76 (95% CI: 5.33, 6.20) days vs. 4.66 (95% CI: 4.62, 4.69) days].

3.4. Current EDC avoidance and MCCs

Fully adjusted estimates (Model #2) of avoidance of select product ingredients with MCL categories in Dataset #1 and mean MCL and within-individual cycle variability in Dataset #2 are summarized in Fig. 5 (effect estimates for all ingredients in both datasets and both Models #1/#2 are available in Supplemental Table S7). Notably, in Dataset #1, avoiding

ingredients such as formaldehyde [OR (95% CI): 0.78 (0.63, 0.96)] or mercury [OR (95% CI): 0.85 (0.70, 1.04)], or avoiding products with a long list of ingredients [OR (95% CI): 0.82 (0.66, 1.00)] was associated with lower odds of very long/irregular cycles; avoiding PFAS [OR (95% CI): 0.65 (0.44, 0.97)] or petroleum propellants [OR (95% CI): 0.65 (0.46, 0.92)] was associated with lower odds of very short cycles. Compared to avoiding no ingredients, avoiding 1–5, 6–10, and ≥ 11 ingredients were associated with 0.68 (95% CI: 0.51, 0.91) times, 0.62 (95% CI: 0.44, 0.87) times, and 0.79 (95% CI: 0.55, 1.13) times lower odds of very long/irregular cycles. In Dataset #2, avoiding ≥ 11 ingredients yielded a lower within-individual variability of 4.64 (95% CI: 4.48, 4.79) days compared to avoiding no [4.73 (95% CI: 4.62, 4.85) days] or 1–5 ingredients [4.81 (95% CI: 4.67, 4.95) days].

3.5. Results from secondary and sensitivity analyses

P-for-interactions of potential effect modifiers are summarized in Supplemental Tables S8–S17, and selected stratified associations (chosen based on the magnitude and biological relevance of the effect differences) are shown in Fig. 6 and Fig. 7. The key findings (with larger differences between effect estimates) related to PCP usage frequency include: higher odds of very long/irregular cycles among high (vs. low) frequency users of body fragrance or semi-permanent dye that were only observed among participants without a PCOS diagnosis, and stronger associations between longer mean MCL and high-frequency usage of 1) dry shampoo, comparing those with fibroids vs. without, 2) body fragrance, comparing those with endometriosis vs. without, and 3) eye makeup, comparing Hispanic vs non-Hispanic White participants. Key findings related to EDC ingredients include: lower odds of very long/irregular cycles for individuals avoiding (vs. not avoiding) BHA, formaldehyde, or phthalates, only observed among the age 18–29 group (when compared to age 30–39 group).

Among hair product users, using fragranced hair oils, touch up coloring, hair growth, or smooth/keratin product were associated with higher cycle variability compared to those who use non-fragranced versions of these products (Supplemental Table S18).

Individuals in the states (detailed in Supplemental Table S2's footnote) without enacted legislation regarding EDCs in PCPs were more prevalent in the very short or very long/irregular cycle groups (51% among those with very short cycles, 48% among those with cycles 21–39 days, and 56% among those with very long/irregular cycles). When stratified by regions, significant effect modifications are observed for frequent lip product or sunscreen use (p-for-interactions in Table S19). An association between frequent lip product use with higher odds of very long/irregular cycles was only observed among participants in the states without enacted legislations regarding EDCs in PCPs [OR (95% CI): 1.32 (1.01, 1.72)], compared to OR (95% CI): 0.30 (0.13, 0.70) among participants in California. Similar regional variations are found for frequent sunscreen use and very long/irregular cycles [OR (95% CI): 1.13 (0.83, 1.54) in other states without legislations vs. OR (95% CI): 0.25 (0.08, 0.76) in California].

Among participants who provided a response to whether attempting to conceive (4,479 participants with 60,687 cycles) in Dataset #2, only 7.8% of the cycles (consisting of 7.4%

reporting trying to conceive naturally, and 0.4% reporting trying to conceive via assisted reproduction technologies) were reported to have been actively attempting to conceive.

As shown in Supplemental Figs. S4 & S5, most co-exposure patterns are weak to moderate. When evaluating overall PCP use patterns reflected by three membership groups showing high, moderate, or low overall use (group composition shown in Supplemental Figs. S6 & S7), PCP use patterns showed little evidence of strong associations with menstrual characteristics: e.g., comparing the participant groups with high vs. low overall past-year PCP use, odds ratios for very long/irregular or very short cycles were small in magnitude, with 95% CI including the null (Supplemental Table S20). From penalized regression approaches (Supplemental Tables S21 & S22), the Elastic Net model for past-year high-frequency use retained a subset of products (semi-permanent hair dye, hair gel, blush/bronzer, nail polish, and hair oil) with penalized odds ratios > 1 for very long/irregular cycles, and these products' individual effect estimates were consistent in direction with the main analyses. Similar consistency was observed for PCPs retained in the model using exposures relating to childhood-initiation. Within moderately correlated groups (e.g., cosmetics, nail products), the penalized regression models tend to retain one key predictor of MCCs, such as blush/bronzer, or nail polish.

Characteristic of those avoiding vs. not avoiding ingredients are shown in Supplemental Tables S23–S24. Sensitivity analyses results when using 32 days as cutoff for long MCL are shown in Supplemental Tables S25–26. Supplemental Fig. S8 illustrates 1) key EDC-relevant PCP use characteristics that were associated with adverse menstrual characteristics or 2) key EDC avoidance behaviors that were associated with more favorable menstrual characteristics, that remained statistically significant after global Benjamini and Hochberg false discovery rate (FDR) adjustments. All FDR-adjusted p-values are shown in Supplemental Table S27.

4. Discussion

This study examined PCP use (current frequency and age at initiation) and EDC avoidance with menstrual cycle length and irregularity among over 6,000 participants contributing either survey data on cycle length or logged cycles data from mobile applications. We found consistent evidence from self-reported MCL data and logged cycle variability data that both current high usage frequency and childhood initiation of body fragrance, dry shampoo, permanent hair dye, perms/relaxers, and self-tanner were associated with higher odds of very long, very short, or irregular cycles. We also found suggestive individual associations for other products such as certain makeup products (e.g., blush/bronzer), nail polish, or skin lightener products either for childhood initiation or current frequent usage on their impact on MCCs, most of which were consistent with findings from secondary analyses using penalized regressions. We also found that avoiding ingredients like petroleum propellants, formaldehyde, lead, mercury, or talc was associated with more regular cycles. Effect modifications were found for some of these findings when stratified by pre-existing gynecological conditions, race/ethnicity, or age groups.

To our knowledge, our study is amongst the first to investigate the direct associations between EDC-relevant PCP usage patterns and menstrual cycle patterns. Our analysis adds behavioral and lifestyle insights to the existing evidence which primarily evaluated specific EDC biomarkers and menstrual health (Buck Louis et al., 2011; Ding et al., 2022; Hammer et al., 2020; Jackson et al., 2011; Jukic et al., 2016; Li et al., 2024b; Lum et al., 2017; Lyngsø et al., 2014; Nishihama et al., 2016; Pollack et al., 2018; Zhu et al., 2019). Several studies have examined EDC-relevant PCP usage in relation to other relevant reproductive outcomes, such as early menarche (James-Todd et al., 2011), hormone-sensitive cancers (Chang et al., 2024), perinatal glycemic control (Bellavia et al., 2019; Preston et al., 2025), preterm birth (Chan et al., 2023b), and uterine fibroids (Ogunsina et al., 2025). These conditions either involve shared mechanisms of hormonal/endocrine disruption (e.g. changes in hormones relevant to the hypothalamic-pituitary–gonadal axis functioning) or represent long-term health outcomes associated with menstrual irregularity (Soria-Contreras et al., 2022).

Our findings of frequent use of body fragrance, hair gel, and nail polish with higher odds of both very short and very long/irregular cycles partially align with the findings of Jukic et al. and Nishihama et al. that high phenols, parabens, or phthalate biomarker concentrations shortened cycle length; these chemicals are commonly found in body fragrance, hair gel, and nail polish (Guo and Kannan, 2013; Johnson et al., 2022; Young et al., 2018). Our findings of childhood initiation of cosmetics such as eyebrow makeup or eyeliner, or current frequent use of blush/bronzer and higher odds of very long/irregular cycles are consistent with findings reported by Zhou et al.'s findings that PFAS being associated with longer cycle length and higher cycle variation, as multiple PFAS chemicals are found in cosmetic products to enhance water resistance and product durability (Commissioner, 2024). For Dataset #2 which uses logged cycle data, to aid interpretation, we note that in a prior study from the same cohort (Li et al., 2023), average within-individual cycle length variability was approximately 4 to 6 days across most subgroups, with mid-reproductive-aged individuals showing values around 3.8 to 4.7 days, which we view as a reference. In our study, some associations had relatively small differences (e.g., body fragrance ≥ 3 days/week vs. never in the past year to a few/year: 4.84 vs. 4.58 days), while others were somewhat larger (dry shampoo childhood initiation vs. adulthood/never: 6.16 vs. 4.73 days). Overall, these changes associated with individual PCPs should be interpreted as small-to-moderate, consistent with our expectations that individual PCP exposures are unlikely to cause extremely large disruptions of menstrual regularity compared to other major health or lifestyle factors that we adjusted for in our models.

On the contrary, we observed inconsistent findings when evaluating age at initiation vs. current frequency related to sunscreen use, where younger age at initiation of sunscreen use yielded higher cycle variability, but frequent current use yielded lower cycle variability. Possible factors warranting future investigation include 1) residual confounding of sociodemographic factors beyond self-reported SES and education level that may impact sunscreen usage or access to safer products, 2) residual confounding of physical activity beyond a one-time self-reported category, particular outdoor activity that can be correlated with sunscreen use (Mínguez-Alarcón et al., 2019), 3) lack of data on individual-level ingredients of sunscreen used (i.e., whether it was mineral sunscreen or contained chemical

UV filters), and 4) lack of data on individual level vitamin D levels (likely correlated with season, time outdoors, and sunscreen usage), as studies have shown that vitamin D deficiency may be a risk factor for long/irregular cycles (Jukic et al., 2018, 2015).

Our findings of lower odds of very short or very long/irregular cycles and reduced cycle variability among those avoiding certain chemicals (such as petroleum propellants, formaldehyde, or PFAS) and heavy metals (like lead or mercury) when purchasing PCPs offers potential actionable steps to promote menstrual health. In the US, federal regulations on EDCs in cosmetic products remain limited, with variations in state-level policies such as California's Proposition 65 requiring warning labels for products containing carcinogens or reproductive toxicants since 1986 (OEHHA, 2016), and some other states recently enacted to regulate chemicals (such as PFAS) in cosmetics in the past 3 years ("Reforming federal cosmetics law," 2023). For heavy metals, although there is some regulation on arsenic, lead, and mercury by the US Food and Drug Administration (FDA), studies still found that certain products (e.g., skin-lightening products) contained mercury levels exceeding the regulation's concentration limit. Additionally, these products may be disproportionately impacting certain ethnic groups and communities (Banala et al., 2023; Bastiansz et al., 2022; Edwards et al., 2023; Hamann et al., 2014; Pollock et al., 2020; Porterfield et al., 2024). Our sensitivity analyses found that the positive associations between certain PCPs (lip products and sunscreens) and long/irregular cycles were observed only among states without these legislations. This suggests that comprehensive EDC regulatory approaches may play an active role in the associations between certain PCPs and MCCs. Confirmation is needed in future studies, especially further evaluating potential confounding by health-promoting lifestyle factors that may correlate with EDC avoidance. Our descriptive analyses suggest that participants who reported avoiding more chemicals tend to have higher SES, healthier lifestyles, and in some cases higher overall PCP use, raising the possibility of residual confounding by socioeconomic or behavioral factors in the observed associations between chemical avoidance and menstrual patterns. Future studies on chemical avoidance should account for these factors, and our findings of chemical avoidance and PCP use as behavioral factors, potentially impacted by multiple individual and social factors, suggest the need for future research and public health policy developments to 1) investigate different impacts across diverse sociodemographic groups, 2) develop evidence-based recommendations for increasing consumer awareness and reducing EDC exposures, and 3) strengthen regulations and monitoring of EDC exposures and address disparities in access to safer PCPs that disproportionately impact vulnerable populations (Boyle et al., 2021; Chan et al., 2023a; Collins et al., 2023; Dodson et al., 2021; Johnson et al., 2022; Mandeville et al., 2024; Martin et al., 2022), which may be ultimately more effective than strategies that rely on individuals (who may have limited ability to recognize and avoid specific ingredients) to manage their own exposure.

Our secondary analyses found evidence suggestive of effect modifications by pre-existing gynecological conditions, age, and race/ethnicity. Participants with specific health conditions like fibroids or endometriosis had stronger associations between frequent use of certain PCPs and long MCL than those without these conditions, while the reverse was observed for individuals with vs. without PCOS, highlighting the need to further investigate potential biological interactions between estrogenic vs. androgenic EDCs in these PCPs and estrogen-

sensitive vs. androgen-excess types of gynecological conditions. However, we acknowledge the possibility that some individuals with PCOS may have been on medications (e.g., metformin) that may influence cycle length, which may bias the estimates towards the null. Although we included individuals with pre-existing gynecological conditions such as endometriosis, PCOS, and fibroids in order to be inclusive of those affected by menstrual disorders, our exclusion of hormone users removed a subset of participants, likely those with more severe symptoms who are most likely to be using hormones for symptom management. As a result, the generalizability of our results to all individuals with these conditions, especially severe cases, may be limited. We also found age-related differences where younger (18–29 years) participants appeared to be more heavily impacted by not avoiding certain chemicals than older participants. Generational variations in awareness of EDCs (Pravednikov et al., 2024) and differences in hormone sensitivity across multiple windows over the reproductive life course (Chow and Mahalingaiah, 2016; Delbes et al., 2022; Peebles and Mahalingaiah, 2024) warrant future investigation.

Limitations of the study include potential misclassification and recall error for the self-reported MCL category. However, we expect this to be non-differential, as participants respond to the question related to their usual MCL on a separate baseline survey different from the PCP surveys. In addition, the logged cycles data from mobile applications are unlikely to be impacted by PCP survey responses, although residual error may be possible for logged cycles data if skipped logging was not adequately identified (Duttweiler et al., 2024). We expect possible misclassification of the one-time, self-reported PCP variables and EDC avoidance patterns, and current data available in this study does not support an accurate investigation of lifetime cumulative risk. Of note, the age at initiation variable should be interpreted only as an indicator of early vs. later onset of use, rather than a measure of cumulative exposure. Future studies with more detailed life course PCP use history are needed to better characterize cumulative exposure and to assess whether any effects of early initiation persist or “wear off”. For some associations (e.g., hair growth products and irregular cycles), reverse causation may be possible due to hair loss being a common symptom of PCOS. Unmeasured or residual confounding, especially on early-life developmental factors, may exist for the associations between childhood initiation of certain PCPs and MCCs. Multiple testing across individual exposure-outcome associations might increase the type I error rate. However, as we are among the first to evaluate the associations between PCP usage, EDC avoidance, and menstrual health, our findings are presented as exploratory and hypothesis generating. Our sensitivity analyses using FDR adjustments further identified two robust categories of findings 1) EDC-relevant PCPs that were consistently associated with more irregular cycles, including eye makeup, body fragrance, dry shampoo, permanent hair dye, hair growth products, and perms/relaxers, and 2) EDC ingredients associated with more regular cycles when avoided, including petroleum propellants, formaldehyde, lead, mercury, and talc. These significant findings may help highlight the most concerning PCPs or the most beneficial EDC avoidance actions, while other suggestive, individual exposure-outcome associations may still inform hypotheses and warrant validation in future studies. This work will inform future studies designed to investigate more complex PCP patterns as a source of EDCs and their impact on menstrual health. As reflected by our secondary analyses, by summarizing PCP co-exposure

patterns into three classes of participants, our cluster-based approach reduced dimensionality while capturing the correlation structure among products; however, we did not find strong evidence of the associations between broad “overall use” of PCPs and MCCs, suggesting that the individual effect of specific products (e.g., those showing consistent evidence across our main models and the Elastic Net models) are more informative as potential hypothesis-generation findings for future research or targeted interventions on specific products of greater relevance to endocrine disruption. Other unmeasured confounders may exist in this analysis. We acknowledge that pregnancy planning behaviors may affect cycle logging patterns, particularly in preconception cohorts or those specifically enrolling pregnancy planners, where menstrual tracking behavior during attempts to conceive can lead to differential cycle data contribution. However, in our analytical dataset from a cohort in which both pregnancy planners and non-planners can enroll, the overall prevalence of active pregnancy planning is relatively low, and only few participants reported use of assisted reproductive technologies during their logged cycles. This suggests that any such impact on cycle characteristics is likely modest. In this context, our findings may be more generalizable to those not engaged in fertility treatment or active pregnancy planning, while being less informative regarding the potential role of PCPs in menstrual patterns among subfertile and fertility treatment-seeking populations. While we conducted secondary analyses comparing fragranced vs. non-fragranced hair products (a product type commonly reported in the literature with fragrance as ingredients and related health impacts) (Helm et al., 2018; James-Todd et al., 2021; Preston et al., 2021; Schildroth et al., 2024), we did not collect this information for other product categories in order to limit participant survey burden; future studies that systematically assess fragrance content across a broader range of PCPs are needed. Finally, as this study includes Apple iPhone users, our results may not be fully generalizable to all menstruating US individuals.

Despite the limitations, our study has several unique strengths. First, we utilized a comprehensive methodology combining self-reported survey data with longitudinal digital cycle logging, which allowed for more precise and complementary menstrual health characterization including both MCL categories and within-individual variability as an indicator of menstrual irregularity. Our study design enabled examination of PCP usage both during childhood and adulthood, providing insights into the impact of both childhood initiation and current frequent usage, two important time windows over one’s reproductive life course. The large, geographically and demographically diverse sample with > 61,000 logged cycles enhanced the generalizability of our findings. By focusing on both PCP usage and EDC avoidance, we aimed to disentangle potentially modifiable exposure sources of EDC, offering a novel angle to understanding risk factors of adverse menstrual health outcomes that may inform future targeted and actionable intervention studies. Furthermore, our analysis explored potential effect modifications, providing insights into identifying groups of individuals who may be more heavily impacted by exposure to certain PCPs or chemicals. These innovations contribute to a more comprehensive understanding of environmental influences on menstrual health and provide a foundation for future targeted research.

5. Conclusions

In conclusion, our study provides novel insights into the associations between personal care product usage, endocrine disrupting chemical avoidance, and menstrual cycle length and variability. We found that individuals with certain pre-existing conditions and younger individuals reported more of these effects. Our findings highlight the critical need for further research and public health interventions that address personal care products, a source of chemical exposures, and their potential reproductive health consequences.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgements

We would like to thank the study participants for consenting and contributing to the advancement of women's health research. We would also like to acknowledge Harvard T.H. Chan School of Public Health staff Carrie Sarcione, Erin Dracup, and Eliana Huffman for their work in supporting this study.

Funding statement

This study received funding from Apple Inc. The funding source provided platforms and software for the collection and management of the data, and participated in the review of the manuscript. The funding source played no role in the design and conduct of this analysis, interpretation of the data, preparation of the manuscript, nor in the decision to submit the manuscript for publication. Co-authors from Apple (K.A. and S.H.S.) reviewed the manuscript for use of name and accuracy, and contributed to revising/editing the manuscript. This research was supported, in part, by the intramural research program of the National Institute of Environmental Health Sciences under award number Z01ES103333. Support for A.M.J. and A.J.W. was provided by the Intramural Research Program of the National Institute of Environmental Health Sciences, National Institutes of Health.

Data availability

Aggregated data that support the findings of this study may be available upon reasonable request from the corresponding author & senior author. Any request for data will be evaluated and responded to in a manner consistent with legal obligation and policies intended to protect participant confidentiality, language in the Study protocol, and informed consent form.

Appendix A.: Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2026.110260>.

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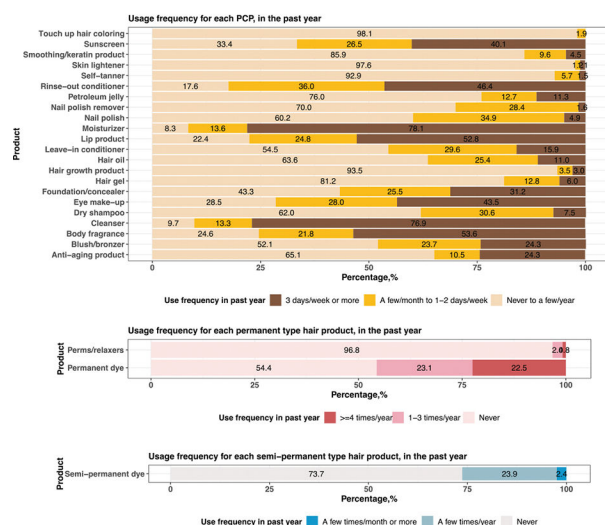
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A. PCP use frequency in the past year



B. PCP age at initiation

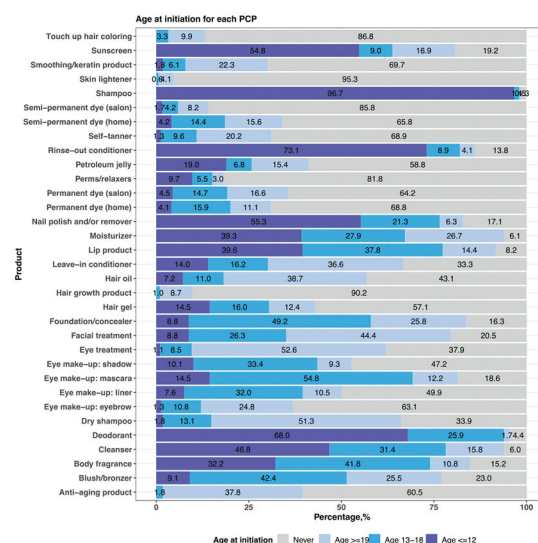


Fig. 1.

Percentages of select PCP (A) usage frequency and (B) age at initiation in Dataset #1. Full distributions of all PCPs in Datasets #1 and #2 are available in Supplemental Tables S3 and S4. Certain groups collapsed due to too few participants reporting usage of 3 days/week or more (touch up coloring: a few/month to 1–2 days/week and 3 days/week or more shown in aggregate), or too few participants reporting initiating the product at age ≤ 12 years (skin lightener, anti-aging product, touch up coloring, and hair growth products: age ≤ 12 and 13–18 years shown in aggregate).

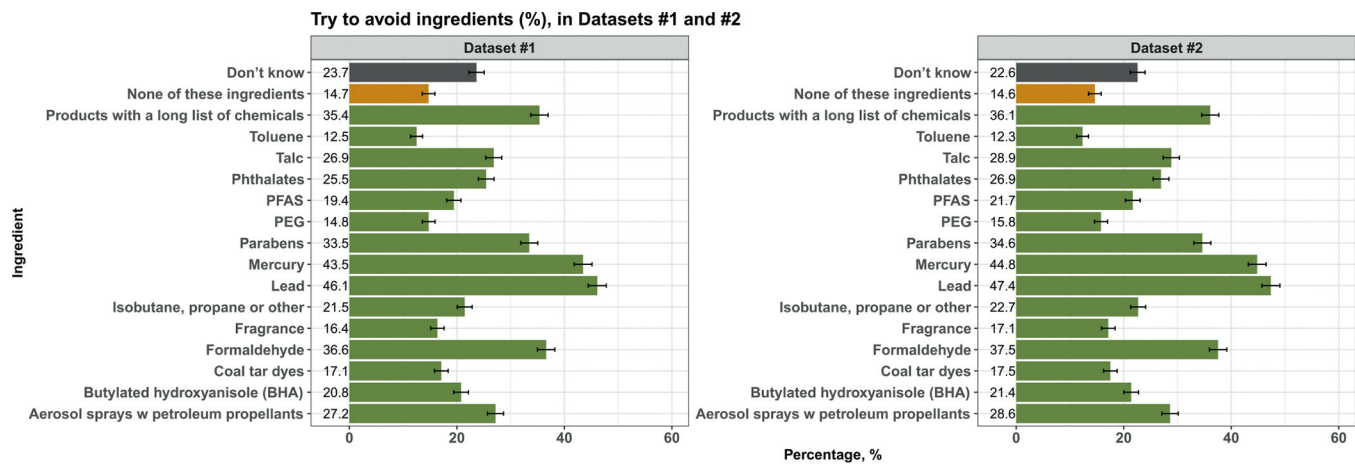
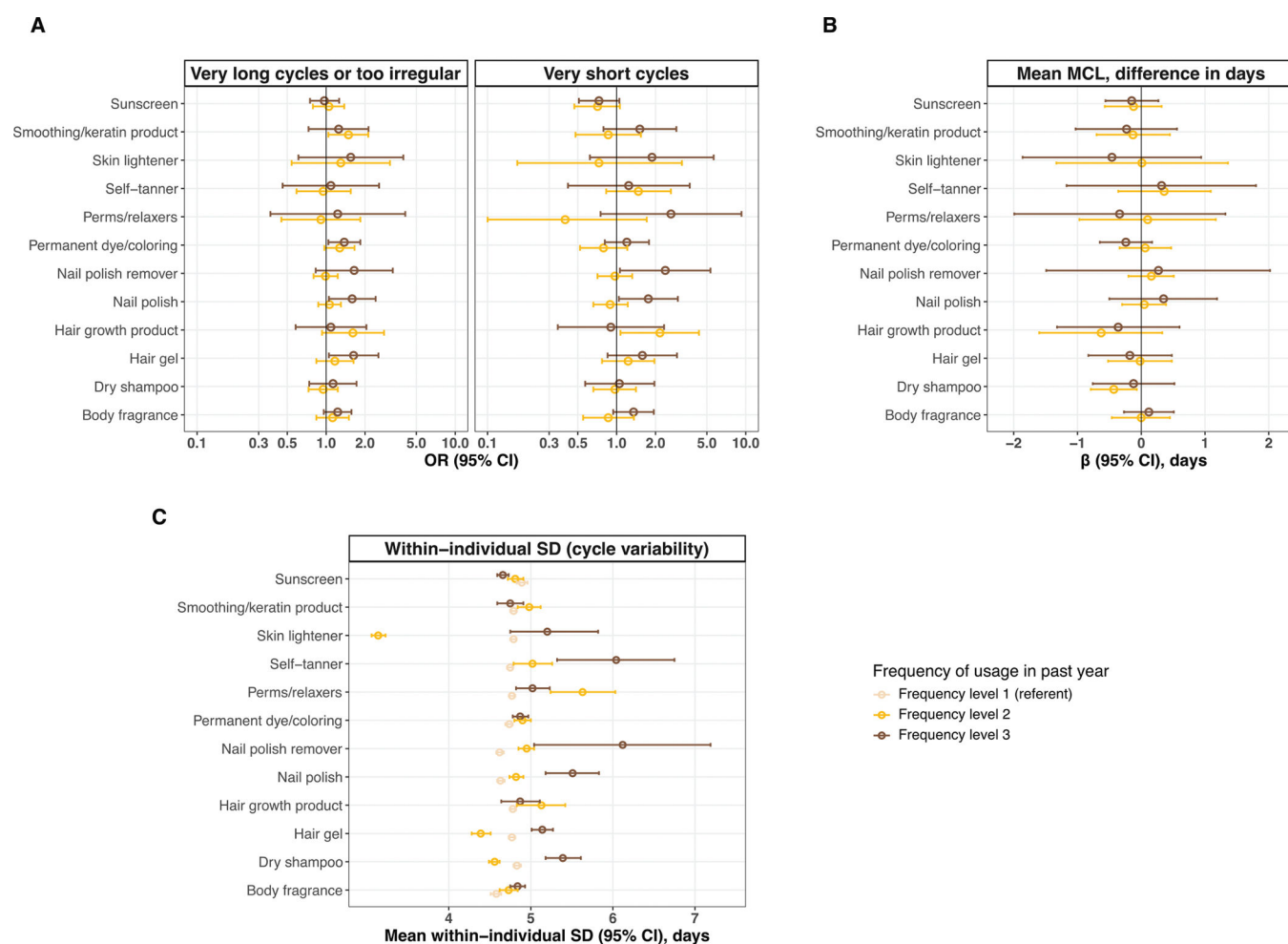
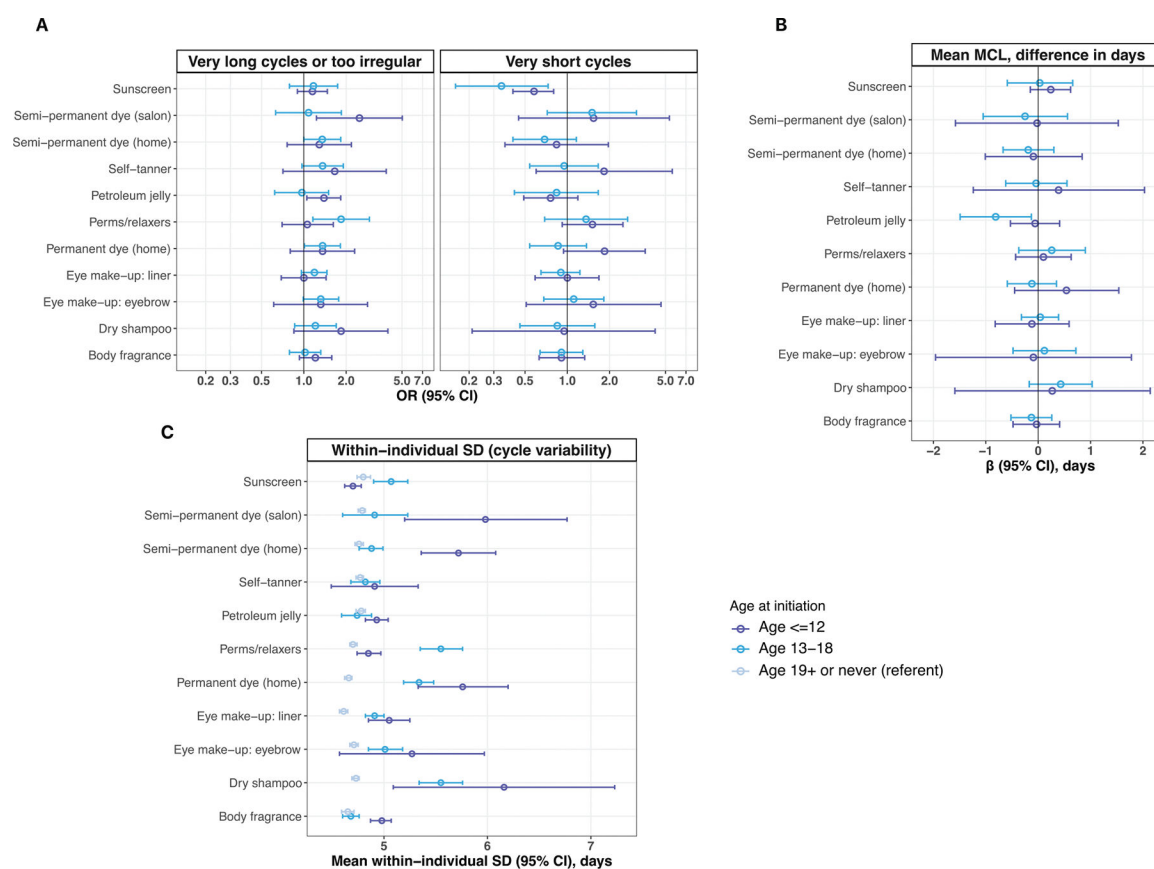


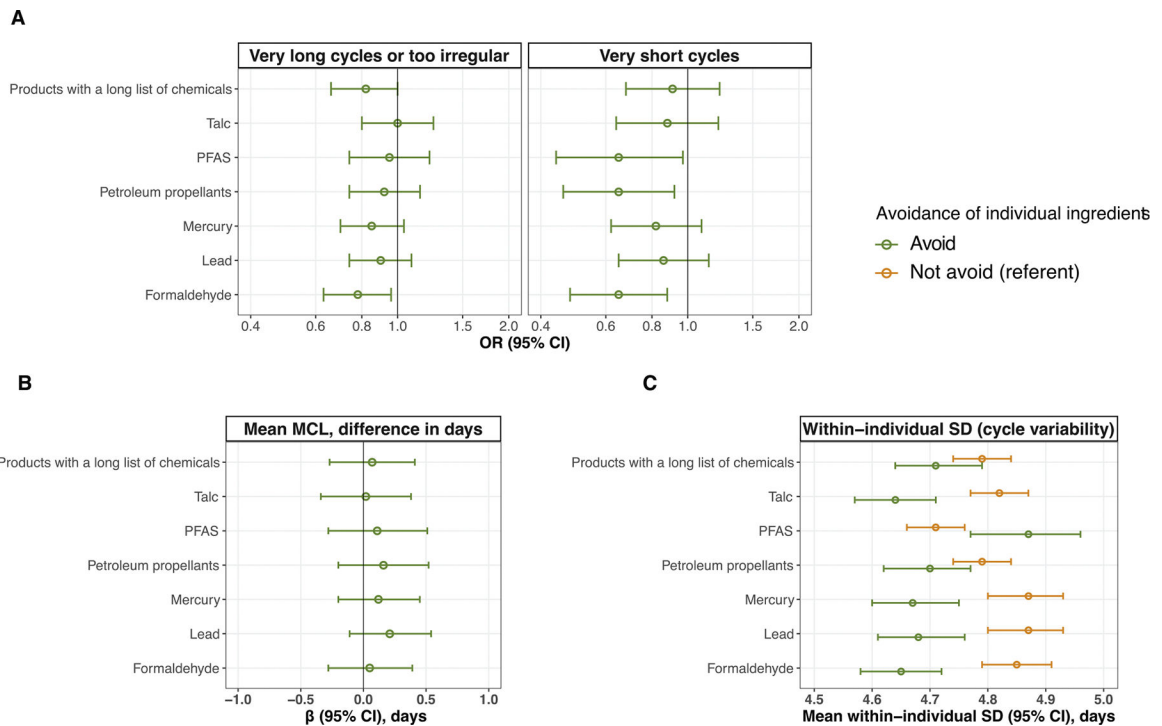
Fig. 2.
Percentages (95% CIs) of avoiding each product ingredient in Dataset #1 and Dataset #2.

**Fig. 3.**

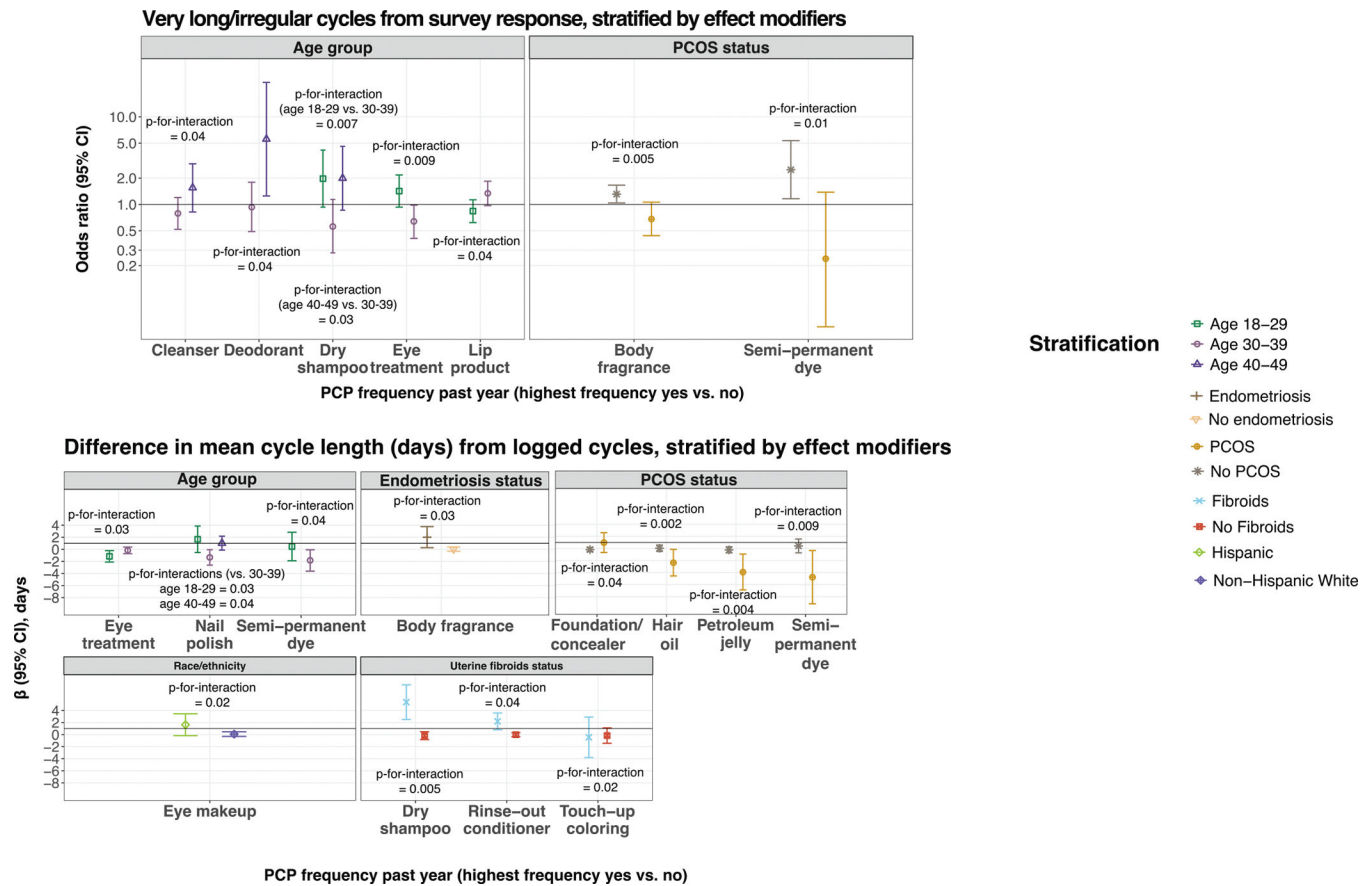
Covariate adjusted effect estimates of associations between select PCP usage frequency and (A) MCL categories in Dataset #1, (B) mean MCL from logged cycles in Dataset #2, and (C) within-individual cycle variability in Dataset #2. Effect estimates for the full list of PCPs in both datasets and both Model #1 and #2 are available in Supplemental Table S5. For most PCPs, frequency groups include: never in the past year to a few/year (level 1), a few/month to 1–2 days/week (level 2), ≥ 3 days/week (level 3); for permanent dye or perms/relaxers, frequency groups include: never in the past year (level 1), 1–3 times/year (level 2), 4 + times/year (level 3).

**Fig. 4.**

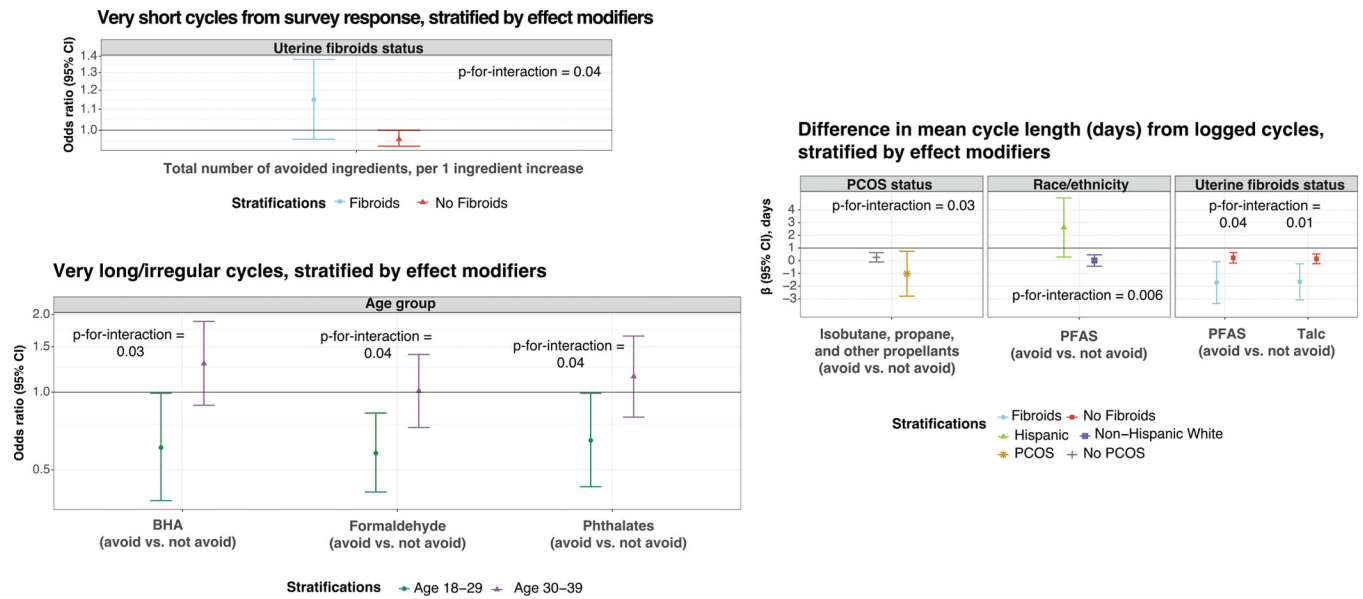
Covariate adjusted effect estimates of associations between select PCP age at initiation and (A) MCL categories in Dataset #1, (B) mean MCL from logged cycles in Dataset #2, and (C) within-individual cycle variability in Dataset #2. Effect estimates for the full list of PCPs in both datasets and both Model #1 and #2 are available in Supplemental Table S6.

**Fig. 5.**

Covariate adjusted effect estimates of associations between individual product ingredient avoidance and (A) MCL categories in Dataset #1, (B) mean MCL from logged cycles in Dataset #2, and (C) within-individual cycle variability in Dataset #2. Effect estimates for the full list of PCPs ingredients in both datasets and both Model #1 and #2 are available in Supplemental Table S7.

**Fig. 6.**

Covariate adjusted effect estimates of associations between current PCP usage frequency and MCCs, stratified by significant effect modifiers. P-for-interactions for testing all potential effect modifiers are shown in Supplemental Tables S8–S12.

**Fig. 7.**

Covariate adjusted effect estimates of associations between EDC avoidance and MCCs, stratified by significant effect modifiers. P-for-interactions for testing all potential effect modifiers are shown in Supplemental Tables S13–S17.

Table 1

Characteristics of the 4155 AWHs participants with relevant survey-based data and 4482 AWHs participants with relevant logged cycle data.

Baseline characteristics (at enrollment) ^a	Participants in Dataset #1:				Participants in Dataset #2: with logged cycles
	Overall	Self-reported usual MCL, days			Overall
		Short (<21 days)	21–39 days	Long (40 + days) or too irregular	
N	4155	329	3088	738	4482
Age in years					
Mean ± SD	33.6 ± 8.3	34.7 ± 8.1	34.0 ± 8.2	31.4 ± 8.3	34.4 ± 8.1
Median (IQR)	34 (27 – 40)	35 (28 – 41)	35 (27 – 41)	31 (24 – 37)	35 (28 – 41)
Age in years, n (%)					
18–19	165 (4.0)	12 (3.6)	113 (3.7)	40 (5.4)	144 (3.2)
20–29	1249 (30.1)	87 (26.4)	872 (28.2)	290 (39.3)	1174 (26.2)
30–39	1560 (37.5)	125 (38.0)	1167 (37.8)	268 (36.3)	1788 (39.9)
40–49	1181 (28.4)	105 (31.9)	936 (30.3)	140 (19.0)	1376 (30.7)
Race and ethnicity, n (%)					
Asian	140 (3.4)	9 (2.7)	101 (3.3)	30 (4.1)	161 (3.6)
Hispanic	266 (6.4)	24 (7.3)	188 (6.1)	54 (7.3)	317 (7.1)
Non-Hispanic Black	204 (4.9)	26 (7.9)	141 (4.6)	37 (5.0)	233 (5.2)
Non-Hispanic White	3074 (74.0)	229 (69.6)	2324 (75.3)	521 (70.6)	3293 (73.5)
Multiple/other races ^b	454 (10.9)	37 (11.2)	324 (10.5)	93 (12.6)	463 (10.3)
Subjective SES scale, n (%)					
Low (0–3)	1202 (28.9)	121 (36.8)	821 (26.6)	260 (35.2)	1094 (24.4)
Medium (4–5)	1806 (43.5)	136 (41.3)	1355 (43.9)	315 (42.7)	1921 (42.9)
High (6–9)	1138 (27.4)	69 (21.0)	908 (29.4)	161 (21.8)	1461 (32.6)
Education level, n (%)					
High school or below	677 (16.3)	95 (28.9)	418 (13.5)	164 (22.2)	561 (12.5)
Some college or tech school	1230 (29.6)	102 (31.0)	877 (28.4)	251 (34.0)	1239 (27.6)
College degree	1271 (30.6)	79 (24.0)	984 (31.9)	208 (28.2)	1486 (33.2)
Graduate school	970 (23.3)	51 (15.5)	806 (26.1)	113 (15.3)	1186 (26.5)
BMI in kg/m ² , n (%)					
Underweight (<18.5)	133 (3.2)	12 (3.6)	100 (3.2)	21 (2.8)	130 (2.9)
Healthy weight (18.5–24.9)	1385 (33.3)	86 (26.1)	1088 (35.2)	211 (28.6)	1515 (33.8)
Overweight (25.0–29.9)	1034 (24.9)	78 (23.7)	805 (26.1)	151 (20.5)	1098 (24.5)
Obesity 1 (30.0–34.9)	639 (15.4)	60 (18.2)	464 (15.0)	115 (15.6)	737 (16.4)
Obesity 2 (35.0–39.9)	457 (11.0)	46 (14.0)	307 (9.9)	104 (14.1)	470 (10.5)
Obesity 3 (≥40.0)	420 (10.1)	35 (10.6)	269 (8.7)	116 (15.7)	450 (10.0)

AWHS: Apple Women's Health Study; BMI, body mass index; SD: standard deviation; SES: socioeconomic status; IQR: interquartile range; MCL: menstrual cycle length; PSS-4: Perceived Stress Scale-4.

^aAll characteristics are based on participant self-report. Numbers may not add to total N (and percentages may not add to 100%) due to missingness. Missingness rates were low (0%–3.1%) for all covariates in both analytical datasets (the only exception is alcohol use among the 4482 participants with relevant logged cycle data, with a missingness rate of 8.4% for this variable within this subset).

^b“Other” included: American Indian or Alaska Native, Middle Eastern or North African, Native Hawaiian or Pacific Islander, or none of these categories can fully describe the participant. Multiple races correspond to those who selected more than one race/ethnicity category.

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